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RESEARCH ARTICLE

Unravelling the spatial and temporal variability of natural disturbances in European forests

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Abstract

1. Despite disturbances intensifying across European forests, it is not well known how their baseline regimes and trends vary across regions and disturbance agents.
2. Using a Landsat-derived dataset of over 3.8 million fire, wind and bark beetle disturbance events (1985–2023), we applied Gaussian finite mixture models to characterise natural disturbance regimes by causal agent, based on disturbance size, severity and frequency. We then analysed their variability across 50-km grid cells and biomes (boreal, temperate, Mediterranean), as well as their long-term trends and transitions across three periods (1985–1997, 1998–2010, 2011–2023).
3. Fire regimes exhibited a marked north–south latitudinal gradient: *severe and rare* fires occurred in boreal and temperate biomes, whereas *moderate* and *large and frequent* fires predominated in Mediterranean and temperate–Mediterranean transitional forests. For wind and bark beetle disturbances, we identified four distinct regimes: *moderate* disturbances predominated in boreal and temperate biomes; *mild and rare* disturbances occurred in Mediterranean and temperate–Mediterranean forests; and *frequent and small*, as well as *severe and large* disturbances predominated in Central Europe and the British Isles, respectively.
4. Approximately 15% of all grid cells shifted to a different disturbance regime during the study period. Fire frequency and severity declined in temperate and Mediterranean forests between the first and second period (1985–1997 and 1998–2010). Wind and bark beetle disturbances decreased in size in boreal and temperate forests during the same period, followed by increases in disturbance severity in temperate forests between the second and third period (1998–2010 and 2011–2023).
5. *Synthesis and applications:* The heterogeneity of natural disturbance patterns and high temporal variability across European Forest suggest the need of adopting context-dependent management strategies tailored to both the dominant disturbance agent and local environmental conditions, particularly in the most vulnerable regions, such as the Iberian Peninsula—with large and frequent fires—and Central Europe—with increasing bark beetle outbreaks.

[Correction added on 3 April 2026 after first online publication: The affiliation for Cornelius Senf has been corrected.]

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KEYWORDS
agent-specific patterns, disturbance regimes, disturbance trends, European forests, fire disturbances, natural forest disturbances, remote sensing, wind and bark beetle disturbances

1 | INTRODUCTION

Natural forest disturbances, such as fires, wind-throw and insect outbreaks, are key ecological processes shaping forest dynamics and structure over time (Asner, 2013; Kashian et al., 2022; Pickett & White, 1985). These events influence the functioning of forests ecosystems and the services they provide in multiple ways across space and time (Knoke et al., 2021; Lecina-Diaz et al., 2024; Pereira et al., 2021). For instance, disturbances can affect forest carbon storage by altering forest coverage and structure (Mack et al., 2021; Pugh et al., 2019), which is crucial for mitigating climate change (Pan et al., 2024). Wildfires alone released $0.4 \pm 0.1\,PgC\,year^{-1}$ of carbon in northern boreal and temperate forests between 2001 and 2021, offsetting nearly 18% of the region's total net carbon sink (O'Sullivan et al., 2024). Beyond carbon dynamics, forest disturbances also affect biodiversity (Thom & Seidl, 2016; Viljur et al., 2022), sometimes promoting species and functional diversity through canopy opening (e.g. in ground-layer plants and soil microbial communities), but in other cases leading to biodiversity loss and habitat degradation (Hingston & Grove, 2010; Lóšková et al., 2013).

There is growing evidence that forest disturbance regimes worldwide are undergoing rapid and unprecedented changes (McDowell et al., 2020). Natural disturbances such as wildfires and bark beetle outbreaks have intensified and expanded into regions where they were historically rare or absent (Altman et al., 2024; Potapov et al., 2025), and this increasing trend is expected to continue under ongoing climate warming (Seidl et al., 2017). In addition, disturbance dynamics show considerable heterogeneity, driven by regional differences in environmental conditions, forest management legacies and the causal agents involved (Altman et al., 2024; Seidl et al., 2014; Sommerfeld et al., 2018). This heterogeneity highlights the need to characterise disturbance regimes (i.e. the spatial and temporal characteristics of disturbances occurring over long periods (Turner, 2010); and defined here by the average size, severity and frequency of disturbances) and to assess their trends across regions and agents. Such characterisation can support the identification of regions undergoing regime shifts or approaching ecological tipping points (Turner & Seidl, 2023), improve our understanding of how broad-scale drivers such as climate change interact with local factors (Sommerfeld et al., 2018) and enhance the representation of disturbances in global vegetation models (O'Sullivan et al., 2024; Weng et al., 2012). Ultimately, this knowledge can inform large-scale forest risk assessments and guide adaptive management strategies.

European forests span a wide bioclimatic gradient, ranging from water-limited Mediterranean to temperature-limited boreal regions, resulting in a pronounced spatial variability of natural disturbance regimes (de Rigo et al., 2016; Patacca et al., 2022). Wind stands as the largest contributor to natural disturbances in the continent, accounting for 46% of the total damage (Patacca et al., 2022) and predominating in temperate and boreal latitudes (Senf & Seidl, 2021). Both the prevalence of wind disturbance (i.e. the proportion of the total area affected) and the absolute area affected have increased over the past few decades (Senf & Seidl, 2021). Fire accounts for 24% of total damage and predominates in southern Europe (Köhl et al., 2020; Patacca et al., 2022); however, burned area has remained stable and even declined since the 1980s, primarily due to enhanced fire suppression policies and improved detection and response systems (Turco et al., 2016). Bark beetle infestations account for 17% of total damage and mostly affect temperate regions (Hlásny et al., 2021; Patacca et al., 2022). Their impact has intensified over the past 20 years, posing now a major socio-ecological challenge in parts of Central Europe (Blicharska et al., 2020; Hlásny et al., 2021; Patacca et al., 2022). Despite the solid understanding of the role of natural disturbances on European forests and their broad regional patterns (e.g. by biomes and ecoregions; see Patacca et al., 2022; Schelhaas et al., 2003; Seidl et al., 2014), we still lack a standardised, spatially explicit characterisation of dominant disturbance regimes and their spatiotemporal variability across causal agents.

Natural disturbances are inherently stochastic and sensitive to climatic variability, making it challenging to assess their regimes and trends on a continental scale (Johnstone et al., 2016; Seidl & Senf, 2024). Historical ground-based information is crucial for understanding the dynamics of forest disturbances (Patacca et al., 2022), yet these data are frequently scattered, unharmonised and inaccessible to the public (Adame et al., 2022). Important efforts have been made to harmonise continental-scale disturbance records in Europe (see Caudullo & Barredo, 2019; Patacca et al., 2022; Schelhaas et al., 2003). However, this information is often biased towards certain countries and incomplete, as the reported disturbance events are often anecdotal or catastrophic, with many smaller events remaining unreported (Patacca et al., 2022). International forests monitoring networks (e.g. ICP forests) and national forest inventories provide valuable and precise information on the effects of disturbances on forests (Rebollo et al., 2024). However, their spatial and temporal continuity is limited because data is gathered periodically at certain plot locations and not continuously over space or time (International Tree Mortality Network, 2025; Ruiz-Benito et al., 2020). The open accessibility of the Landsat archive (Banskota et al., 2014), coupled with enhanced computational capacity (Gorelick et al., 2017) and the increasing availability of field data for model calibration and validation (Miguel et al., 2024), are enabling the development of increasingly accurate data products. These products now capture forest disturbances with great spatial, temporal and thematic detail (e.g. the European Forest Disturbance Atlas by Viana-Soto & Senf, 2025, provide information on timing, severity and causal agents at continental scales since the 1980s). Such databases open-up new opportunities to investigate the

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spatiotemporal variability of natural disturbance regimes with greater precision, at a continental scale and disaggregated by causal agent.

Our aim here is to characterise the main natural disturbance regimes in Europe and assess how they vary spatially and change over time. More specifically, we addressed the following research questions: (1) What are the main disturbance regimes based on size, severity and frequency of events for (i) fire and (ii) wind and bark beetle disturbances?; (2) Do disturbance regimes vary among and within biomes?; (3) Do disturbance regimes and their spatial distribution vary over time? To answer these questions, we analysed data on 3,825,259 disturbance events that occurred in European forests between 1985 and 2023. We then identified the dominant disturbance clusters (i.e. disturbance regimes) for the two agent groups. We quantified the spatial variability of disturbances between and within biomes (boreal, temperate and Mediterranean), dividing the study region into hexagonal grids of 50 km side length. Finally, we compared changes in disturbance clusters over time, dividing the entire period into three and assessing cluster changes from the first to the third period (1985–1997 and 2011–2023), as well as assessed the annual trends from 1985 to 2023.

2 | METHODS

2.1 | Disturbance information

To identify and characterise disturbance events, we used the European Forest Disturbance Atlas version 2.1.1 (European Atlas, henceforth; Viana-Soto & Senf, 2025). The European Atlas provides annual data on forest disturbances at a 30-m spatial resolution for 38 European countries from 1985 to 2023 (Figure 1). Here, we used three layers of the European Atlas: the year of disturbance, the disturbance severity and the disturbance agent.

The year of disturbance indicates when a pixel experienced a complete or partial loss of forest canopy due to natural or human causes. Disturbances affecting the same pixel in different years are identified independently when a minimum interval of 3 years separates two events; otherwise, only the first event and root agent is considered. However, when disturbances occur in consecutive years, their effects may spread in later years to contiguous pixels that were previously undisturbed. In such cases, these adjacent pixels can be identified as disturbed in successive years. The disturbance severity

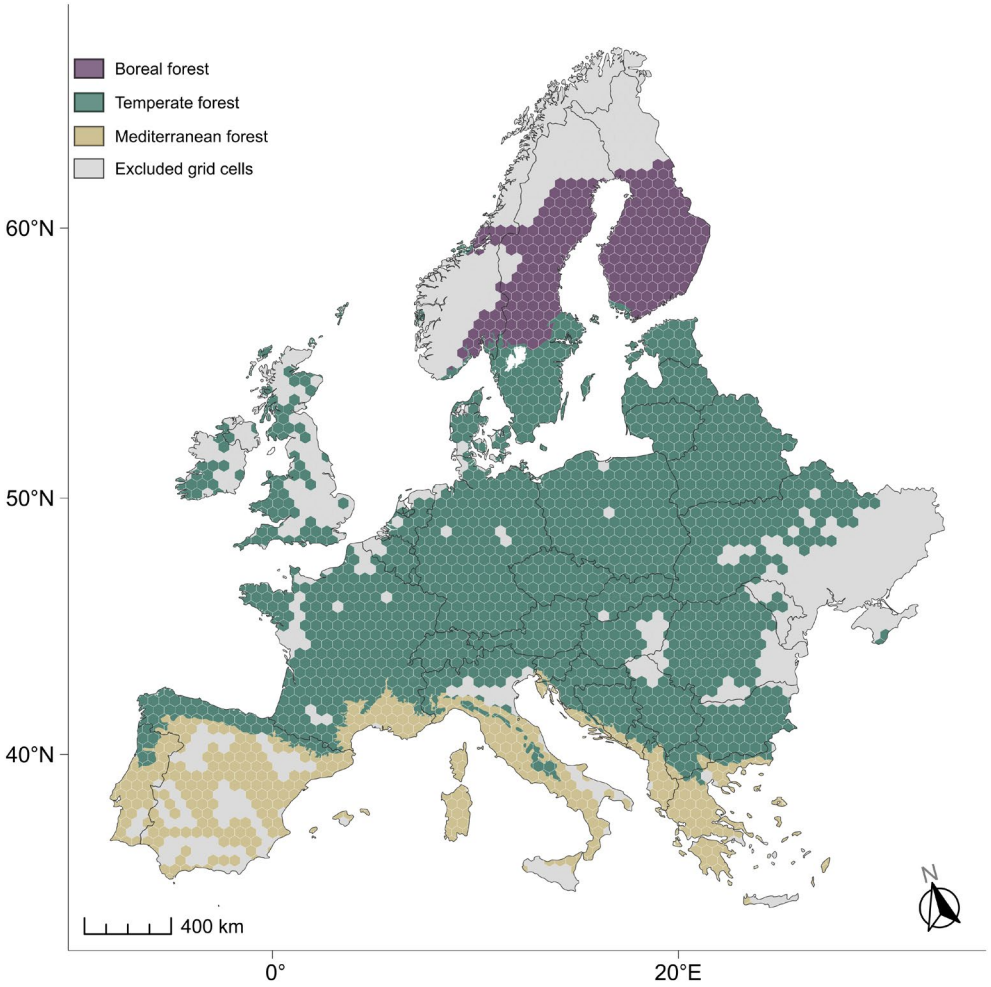


FIGURE 1 The study region showing the hexagonal grid cell division and biomes (boreal, temperate and Mediterranean, based on Olson et al. (2001)) considered for the analyses.

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layer is based on the spectral change in the Normalised Burn Ratio of each disturbed pixel relative to pre-disturbance conditions (ΔNBR), with theoretical values ranging from −10,000 to 10,000 (i.e. lower values indicate higher severity). The disturbance agent layer is derived from the year of disturbance layer and distinguishes among harvest, fire and the combined category of wind- and bark beetle-driven disturbances. Because our aim was to characterise natural disturbance regimes, harvest records were excluded from subsequent analyses. Wind and bark beetle disturbances were grouped into a single category in the European Atlas because current remote-sensing methods cannot reliably distinguish between these agents once tree mortality has occurred (Viana-Soto & Senf, 2025). Although this category also considers other agents (i.e. storms, heavy snow and other insects or pathogens), we use here the term *wind and bark beetle* disturbances for consistency with the European Atlas and because most training data correspond to these two agents.

2.2 | Disturbance characterisation

We aggregated 30×30 m pixels from the European Atlas of the same agent and year into individual disturbance patches, considering the 8-neighbour connectivity, using *landscapemetrics* package (Hesselbarth et al., 2019) in R 4.3.1 (R Core Team, 2023). In total, we identified 650,484 fire and 3,174,775 wind and bark beetle disturbance patches. For each disturbance patch, we extracted summary statistics including patch area (No. disturbed pixels ×0.09 ha, corresponding to the area of a single pixel) and patch severity (mean ΔNBR across all disturbed pixels within the patch). ΔNBR observed values were rescaled to a 0–1 range prior to computing severity statistics, with higher values corresponding to more severe disturbances (Figure S1).

To analyse the spatial variability, we aggregated patch-level metrics into hexagonal grid cells with a 50 km side length covering the entire study area (2415 grid cells in total; see Figure 1). This process was conducted separately for fire and for wind and bark beetles. To reduce noise, we removed grid cells that had a small proportion of forest cover (i.e. <10% of forest cover); were located at northern latitudes than 67°N, where false disturbances have been reported (Viana-Soto & Senf, 2025); or were located in tundra or grasslands, because these biomes are dominated by treeless vegetation (Olson et al., 2001). For each grid cell and year, we calculated 20 disturbance attributes (i.e. mean, standard deviation, coefficient of variation and the 50th and 95th percentiles) for disturbance size, severity, spatial frequency and rate. Disturbance frequency represents the number of disturbances per year normalised by the percentage of forest cover within each grid cell (No. events year^{-1}), whereas rate represents the forest area disturbed normalised by the total forest extent in the grid and expressed as an annual percentage (% year $^{-1}$). We then selected three orthogonal disturbance attributes (see Figure S2 and Table S2 for more details) key to capture the dimensions of disturbance regimes: mean disturbance size (ha), mean disturbance severity (scaled from 0 to 1) and disturbance frequency (No. events year^{-1}). All variables were standardised using logarithmic transformation to improve normality and scaled to a range between 0 and 1 to enable comparison across metrics (Altman et al., 2024). To analyse temporal variability, we used the annual data on size, severity and frequency of disturbances, and aggregated them for the following periods: 1985–1997, 1998–2010 and 2011–2023. Size and severity were aggregated using the mean, while frequency was aggregated using the total number of events (No. events period^{-1}). For subsequent analyses, we excluded grid cells with fewer than three disturbances per period. This threshold was set to avoid using small sample sizes to characterise their disturbance regimes, while avoiding excessive data loss resulting from higher thresholds (Figure S3). Throughout the manuscript, we refer to grid cells with fewer than three disturbances per period as residual activity grid cells.

We used statistical and clustering approaches to identify predominant disturbance regimes in Europe by agent group (i.e. fire, and wind and bark beetles) and assess their spatial distribution and temporal changes. This involved first reducing the dimensionality of three disturbance attributes (i.e. size, severity and frequency) through principal component analysis (PCA). This allowed us to establish a common reference space based on the third period, onto which the first and second periods were projected for comparison to understand changing dynamics of disturbances. Using the third period as reference responded to the higher mapping accuracies reported in the European Atlas for these years than prior to the 2000s (Viana-Soto & Senf, 2025). The first and second principal components were then used as input variables in an unsupervised clustering Gaussian finite mixture model (GMM) to identify grid cells with similar forest disturbance regimes. This was performed using *mclust* package (Scrucca et al., 2023). Performing PCA prior to GMM also improved cluster separation and enabled identifying the disturbance attributes that contributed most to cluster differentiation based on their PCA loadings.

We determined the optimal number of clusters to characterise disturbance types using the Bayesian Information Criterion (BIC) and Integrated Complete Likelihood (ICL) (Scrucca et al., 2023). Then, we characterised the resulting disturbance clusters using parametric bootstrapping with 9999 permutations, and computed each cluster's mean, standard deviation, and 5th and 95th percentiles for disturbance size, severity and frequency. We assessed whether there were any significant differences between the clusters using the non-parametric Wilcoxon signed-rank test (Wilcox, 2010). To ensure that the resulting disturbance clusters and their spatial distribution were not dependent on the grid cell size (50 km side length) and the time periods considered, we replicated the analysis using different grid cell sizes (25 and 100 km) and periods (1985–2004 and 2005–2023), and obtained similar results in all cases (Figures S4 and S5).

We assessed the temporal changes in disturbance regimes in two ways. First, we calculated the percentage of grid cells assigned to each cluster per period, as well as the transitions between clusters across periods. Secondly, we estimated temporal trends by calculating the annual averages of the three disturbance attributes using relative Sen's slope (Sen, 1968). This is a non-parametric estimator that is robust to outliers, making it appropriate for minimising the influence of individual extreme disturbance years.

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3 | RESULTS

3.1 | Characteristics of disturbance clusters

The results of the PCA for fires showed that the first component (PC1, explaining 47.1% of the variance) was negatively correlated with severity and positively correlated with frequency. Meanwhile, the second component (PC2, explaining 32.2% of the variance) was negatively correlated with disturbance size (Figure 2a). The GMM identified three distinct disturbance clusters in the PCA space. The largest and most compact of these clusters was centred near zero on both PCA axes, representing intermediate values of size, severity and frequency. We refer to this group as the *moderate* cluster (20.3% of all grid cells considered; Table 1). A second, more dispersed cluster was distributed along the severity loading vector, indicating high severity and low fire frequency in those grid cells (the *severe and rare* cluster, accounting for 7.2% of all grid cells). The smallest cluster was distributed along the frequency and size loading vectors, indicating frequent and large, but less severe, fires (the *large and frequent* cluster, accounting for 5% of grid cells).

The PCA results for wind and bark beetle disturbances showed that the first principal component (PC1, explaining 52.3% of the variance) was negatively correlated with disturbance size and positively correlated with frequency. The second component (PC2, explaining 34% of the variance) was strongly negatively correlated with severity (Figure 2c). The GMM identified four distinct disturbance clusters. The largest cluster, centred near zero on both axes of the PCA, reflected intermediate values of size, severity and frequency (the *moderate* cluster, accounting for 50% of all grid cells; Table 1). A second, smaller but more dispersed cluster was distributed along the size and severity loading vectors, indicating higher

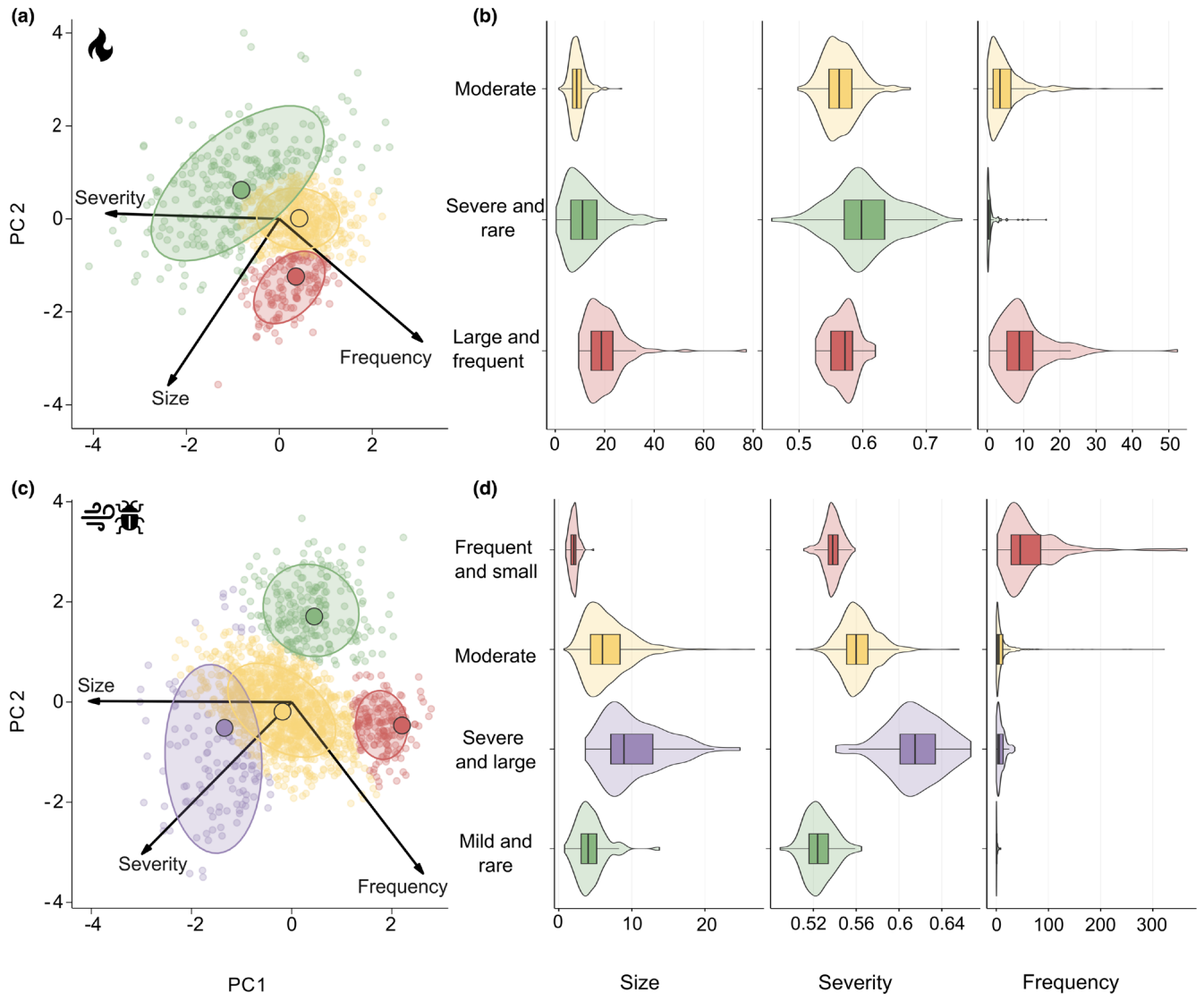


FIGURE 2 Panels (a) and (c) show the first and second axes of the principal component analyses (PC1 and PC2), and the disturbance clusters identified by the Gaussian finite mixture model, based on disturbance size (ha), severity (0–1) and frequency (No. events year^{-1}) for fires (a) and wind and bark beetle disturbances (c) during 2011–2023. Panels (b) and (d) show box and violin plots of the distribution of disturbance attributes across clusters for fires (b) and wind and bark beetle disturbances (d) in the same period.

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TABLE 1 Summary of the disturbance attributes used to characterise fire and wind and bark beetle disturbance clusters from 1985 to 2023. Values are presented as the mean ± standard deviation and the 90% percentile range [minimum, maximum].

Cluster	Size (ha)	Severity (0–1)	Frequency (No. events year^{-1})	Cumulative frequency (No. events period^{-1})	Area disturbed (ha)	Grids (%)
Fire						
Moderate	8.97 ± 2.98 [5.04, 14.32]	0.57 ± 0.03 [0.53, 0.61]	0.49 ± 0.49 [0.08, 1.5]	5.1 ± 5.57 [0.37, 16.91]	1665.68	20.29
Severe and rare	12.87 ± 8.57 [3.57, 29.7]	0.6 ± 0.05 [0.53, 0.7]	0.17 ± 0.22 [0.03, 0.55]	0.93 ± 1.9 [0.08, 3.07]	393.70	7.24
Large and frequent	20.04 ± 8.59 [11.31, 32.3]	0.57 ± 0.02 [0.53, 0.61]	0.88 ± 0.79 [0.19, 2.02]	10.23 ± 7.37 [1.92, 23.94]	8940.94	5.01
Wind and bark beetle						
Frequent and small	2.04 ± 0.55 [1.21, 2.97]	0.54 ± 0.01 [0.52, 0.55]	5.18 ± 4.72 [0.99, 13.24]	64.64 ± 60.87 [12.24, 164.7]	4916.72	7.45
Moderate	6.66 ± 3.13 [2.81, 12.42]	0.56 ± 0.02 [0.54, 0.59]	1.02 ± 1.87 [0.08, 3.31]	11.43 ± 22.8 [0.34, 40.24]	1978.60	50.03
Severe and large	10.3 ± 4.27 [5.3, 17.68]	0.61 ± 0.03 [0.58, 0.65]	0.89 ± 0.79 [0.07, 2.5]	8.42 ± 8.03 [0.2, 24.97]	1421.50	3.44
Mild and rare	4.32 ± 1.94 [1.82, 8.08]	0.53 ± 0.01 [0.51, 0.55]	0.21 ± 0.25 [0.03, 0.74]	1.02 ± 1.47 [0.09, 3.87]	140.54	7.54

values for both attributes (*severe and large* cluster, accounting for 3.4% of all grid cells). In the opposite direction, a third cluster was strongly associated with frequent but smaller disturbances (*frequent and small* cluster, accounting for 7.5% of all grid cells). Finally, the fourth cluster was distributed in the direction opposite to the severity loading vector, indicating neither high frequency nor large size (*mild and rare* cluster, accounting for 7.5% of all grid cells). We found significant differences ($p < 0.05$) between all clusters in terms of size, severity and frequency according to the Wilcoxon test, except between the *moderate* and *mild and rare* frequency regimes for wind and bark beetle disturbances ($p = 0.06$).

3.2 | Main disturbance clusters and their spatial distribution across European forest

The three fire clusters (Table 1; Figure 2a,b) showed clear spatial patterns and marked differences between biomes, with most fires occurring in southern latitudes (Figure 3a). In the Mediterranean biome, two clusters predominated: the *large and frequent* cluster—characterised by its high recurrence and extensive spread—in the western Iberian Peninsula and the *moderate* cluster across the rest of the Mediterranean and Mediterranean–temperate transition forests. In temperate and boreal biomes, most grid cells showed residual activity (67.5% of all grid cells). Nonetheless, *severe and rare* clusters occurred in parts of Sweden, Finland, Belarus and Ukraine, as well as in some Mediterranean grid cells.

The four wind and bark beetle clusters that we identified (Table 1; Figure 2c,d) also exhibited a clear spatial pattern that varied across biomes, with most activity occurring in temperate and boreal regions (Figure 3b). In these two biomes, the *moderate* cluster dominated with three exceptions: Central Europe, where the *frequent and small* cluster emerged; the western British Isles and parts of southern Scandinavia, where disturbances were *severe and large*; and the Mediterranean–temperate transition zones, where the *mild and rare* cluster dominated. In the Mediterranean biome, wind and bark beetle events were mostly residual, although some areas, such as southern France and central Italy, exhibited *mild and rare* regimes.

3.3 | Temporal variation of disturbance regimes

For fires, 15.4% of all grid cells shifted from one cluster to another between the first and third periods (1985–1997 vs. 2011–2023; see Figure 4a; Figure S6; Table S1). Most transitions occurred in temperate and boreal biomes, shifting from *severe and rare* clusters to residual activity (31.7% of all changes; Figure 4a), and in the Mediterranean biome, from *moderate* to *severe and rare* clusters and from *large and frequent* clusters to *moderate* ones (21% and 14.5% of all transitions, respectively; Figure 4a; Figures S7 and S8).

At the biome level, annual trend analysis showed no significant changes in fire size, but significant decreases in fire severity and frequency in temperate and Mediterranean forests between the first and second periods (1985–2010; Table 2). In boreal forests, it is remarkable that a significant decrease in fire frequency between 1985 and 2010 was followed by a steep increase between 2011 and 2023 (Table 2). At the grid-cell level, no clear spatial patterns emerged (Figure 4b–d), except for increases in disturbance severity in parts of Italy and the Balkan Peninsula, and significant decreases in frequency in the central and eastern Mediterranean and the western Iberian Peninsula (Figure 4c,d).

For wind and bark beetles, 16.7% of grid cells shifted between clusters from period 1985–1997 to 2011–2023 (Figure 5a; Figure S6; Table S1). Most transitions were gains in residual activity, mainly from *mild and rare* clusters in Mediterranean latitudes, Sweden and Finland (18.8% of changes), and from *moderate* clusters in temperate latitudes (16.6% of changes). Other common transitions were towards *frequent*

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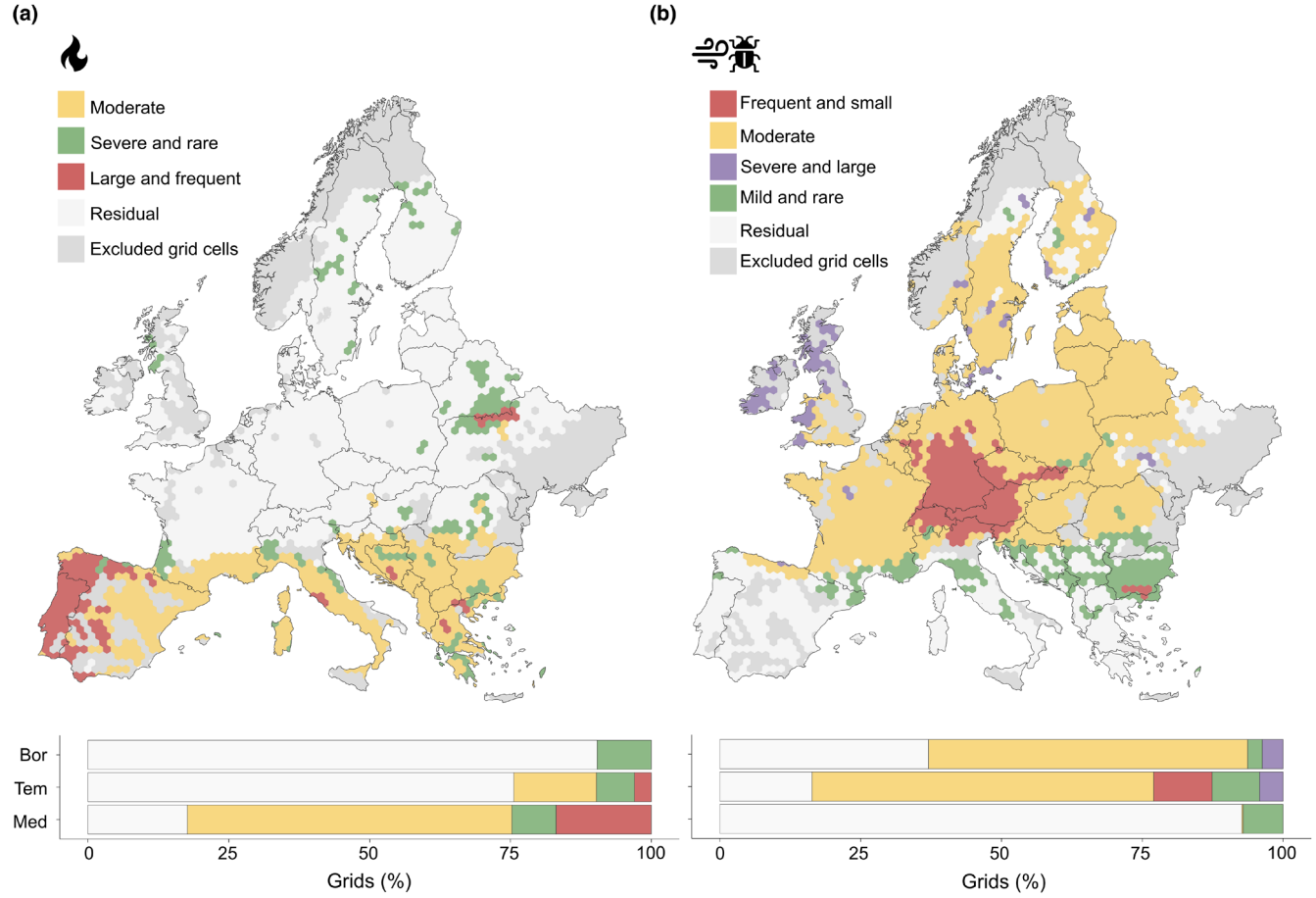


FIGURE 3 Spatial patterns of the identified disturbance regimes caused by fires (a) and wind and bark beetle disturbances (b) across Europe in the period 2011–2023. The bar plots below each map show the proportion (%) of grids within each cluster by biome. Residual activity in light grey shows clusters with fewer than three fire events recorded per period, whereas dark grey clusters were outside the study area.

and small clusters in Central Europe and towards *severe and large* clusters in the British Isles (13.9% and 13.6%, respectively; Figure 5a; Figures S7 and S8). At the biome-level, annual trends showed a significant decline in frequency in the Mediterranean biome and in size in boreal and temperate biomes between 1985 and 2010, along with an increase in severity in temperate forests between 1998 and 2023 (Table 2). At the grid-cell level, trends were heterogeneous. Significant decreases in size and severity occurred in parts of Central Europe, while increases in severity occurred in parts of France, the British Isles and Baltic countries (Figure 5b,c). Meanwhile, significant declines in frequency occurred in Eastern Europe while increases occurred in parts of Central Europe and southern Scandinavian and Baltic regions.

4 | DISCUSSION

4.1 | Characterisation of disturbances in European forests and their spatial distribution

Our results revealed high variability in the spatial and temporal patterns of natural disturbances in Europe, which were effectively captured by the size, severity and frequency of the events. There are multiple metrics to characterise forest disturbances (e.g. area-weighted mean patch size or percentage of forest disturbed, see e.g. Seidl et al., 2020; Sommerfeld et al., 2018; Whitman et al., 2015). However, in our case, the three selected variables proved more effective at differentiating disturbance clusters across time periods and spatial scales than other candidate metrics (e.g. disturbance rate, standard deviation, coefficient of variation, see Table S2 for more details). Furthermore, the fire regimes and spatial patterns we identified were consistent with those reported by Galizia et al. (2021), despite their use of a different set of indicators (i.e. mean annual burned area, fire frequency, percentage of affected area).

Our results show that severity had less variability across clusters and causal agents than disturbance size and frequency, although all groups showed significant differences in severity according to the Wilcoxon test ($p < 0.05$). This limited variability is partly explained by the distribution of ΔNBR values in the European Atlas, where most pixels (98.6%) ranged from −3000 to −1500 before rescaling and from 0.4 to 0.7 after rescaling (see Figure S1 for details). Through the GMM results, we also found that the most common cluster corresponded to disturbances with moderate levels of size, severity and frequency for both fire and the group of wind and bark beetle events. However, our analyses do not capture small-scale mortality gaps involving the death of one or a few trees, as these are usually too small to be detected by Landsat sensors

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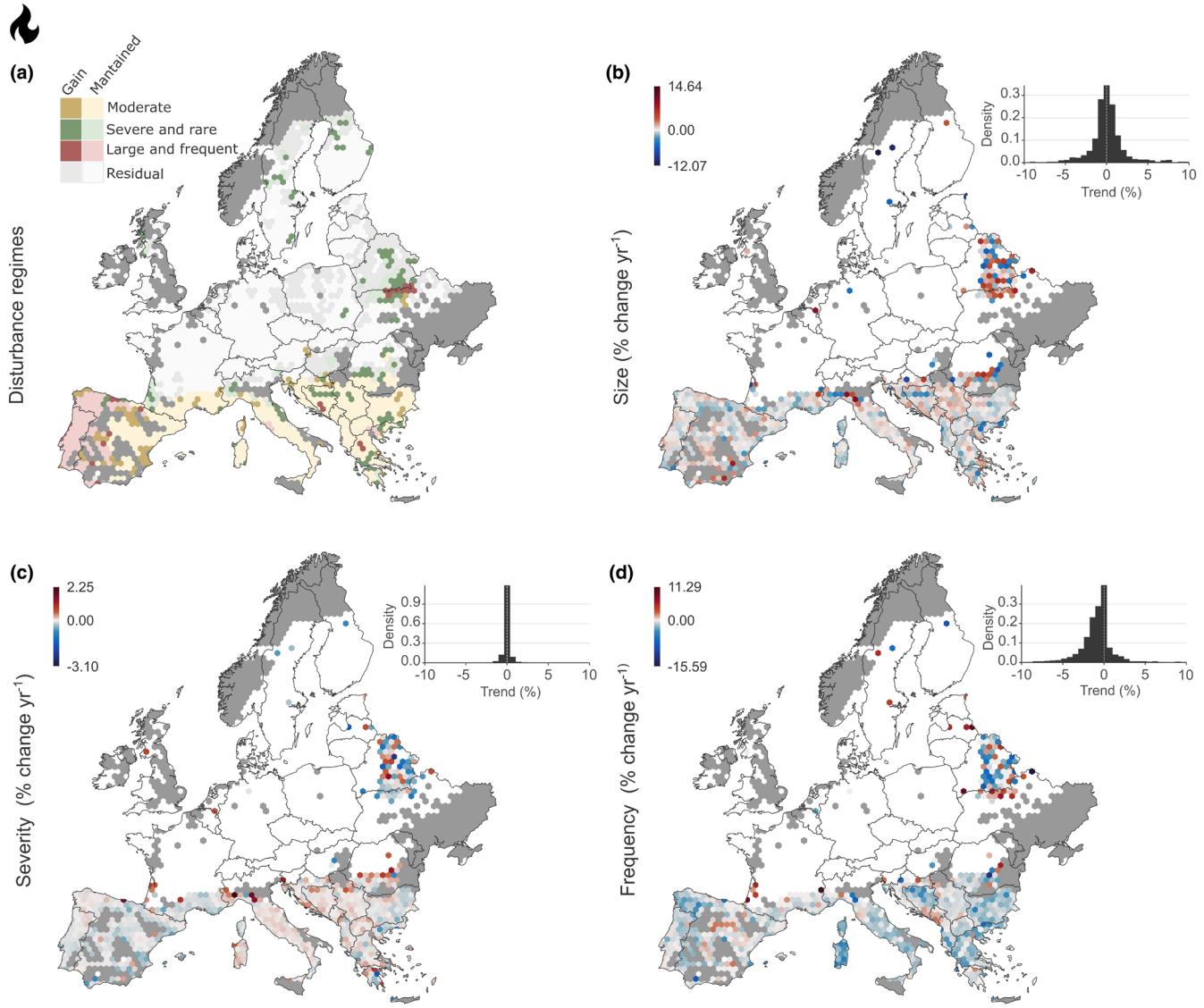


FIGURE 4 Fire disturbance changes through time. Panel (a) shows changes in fire disturbance clusters between periods first (1985–1997) and third (2011–2023). Grids classified as maintained did not change cluster regime, while gain indicates a transition to a different disturbance cluster. Panels (b–d) show relative trends (% change year^{-1}) of properties in Viana-Soto and Senf (2025) of disturbance size (b), severity (c) and frequency (d) from 1985 to 2023. Panels (b–d) display significant trends ($p < 0.05$) colouring the entire grid cells, whereas non-significant trends ($p \geq 0.05$) are shown with coloured points. Grids in grey were those excluded from the analyses.

(Zhu et al., 2022), even though they represent a dominant form of disturbance in many forests worldwide (Muscolo et al., 2014).

We observed little spatial overlap among patches experiencing fires, and wind and bark beetle disturbances, which are the three major natural disturbance agents in Europe (Patacca et al., 2022; Schelhaas et al., 2003; Seidl et al., 2014). Fires primarily occurred in southern Europe, whereas wind and bark beetle disturbances predominated in temperate and boreal regions, as prior work reported (e.g. Patacca et al., 2022; Senf & Seidl, 2021). Nonetheless, these spatial domains converged in the Mediterranean–temperate transition zone, suggesting a higher likelihood of linked or interacting disturbances in this region (Buma, 2015), where one disturbance agent may influence the probability, impact or ecological response of subsequent events (Bigler et al., 2005; Kulakowski & Jarvis, 2013).

Fire regimes exhibited a clear north–south latitudinal gradient, with boreal and temperate forests experiencing mostly residual activity (i.e. less than three fires per period) that increased towards southern latitudes. When fires did occur in northern and central latitudes, they were generally severe and rare. Such high-severity fire regimes have been previously observed in wet and cold environments, where fuel availability is high and ignition conditions (e.g. drought periods) rarely occur (He et al., 2025; Whitman et al., 2015). In contrast, the western Iberian Peninsula exhibited the most extreme fire regime in Europe, characterised by the largest and most frequent fires, in agreement with existing literature (e.g. Galizia et al., 2021; San-Miguel-Ayanz et al., 2012). The extreme fire pattern in western Iberian Peninsula may be influenced by the widespread replacement of natural forests with fire-prone plantations (e.g. *Eucalyptus* sp. and

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TABLE 2 Temporal trends (% change year^{-1}) in disturbance size, severity and frequency for fire and for wind and bark beetle disturbances across Europe per biome (boreal, temperate and Mediterranean), using relative Sen's slope estimates over the entire study period (1985–2023) and two sub-periods (1985–2010, 1998–2023).

Attribute (% change year^{-1})	Level	Fire			Wind and bark beetle		
		1985–2023	1985–2010	1998–2023	1985–2023	1985–2010	1998–2023
Size	Europe	−0.02	−0.12	0.28	-0.17	**−0.59*****	0.15
	Boreal	−0.82	−1.27	−0.3	**−0.95***	**−1.77*****	−0.55
	Temperate	0.1	−0.07	0.35	−0.12	**−0.48***	0.17
	Mediterranean	0.13	0.28	0.28	0.12	0.41	−0.29
Severity	Europe	**−0.06****	**−0.12****	−0.01	**0.02***	−0.02	**0.05***
	Boreal	0.02	0.18	−0.21	0.01	−0.07	0.04
	Temperate	**−0.05***	**−0.14*****	−0.01	0.02	−0.02	**0.04***
	Mediterranean	**−0.06****	**−0.13****	−0.03	0.01	0.04	−0.02
Frequency	Europe	**−0.95***	**−0.89***	−0.92	0.46	−0.12	1.36
	Boreal	1.37	**−2.79***	**4.35*****	−0.76	−1.85	0.6
	Temperate	**−1.35*****	**−1.77***	−1.34	0.44	−0.27	1.35
	Mediterranean	**−1.32***	**−1.74***	−0.71	**−1.41***	−1.73	−0.8

Note: We show significant differences from zero marked in bold.
*$p < 0.05$. **$p < 0.01$. ***$p < 0.001$.

Pinus radiata replacing *Quercus*-dominated forests), which increases fuel continuity and flammability (Cruz et al., 2024). This effect may be further reinforced by the combined influences of climate change (e.g. hotter droughts; Turco et al., 2018), socio-economic and land use changes (e.g. rural abandonment and fuel accumulation; Moreira et al., 2011; Moreno et al., 2014), and direct anthropogenic causes such as arson and negligent burning practices linked to traditional land management (e.g. pasture renewal; Mateus & Fernandes, 2014). The remaining Mediterranean and Mediterranean-temperate transition forests exhibited moderate fire regimes. These regions typically experience recurrent small, cool fires combined with rare but large and intense fires (Archibald et al., 2013; Pausas, 2022), and are associated with hot and dry summers, flammable vegetation (e.g. pine woodlands), rugged terrain and human-related factors (Ganteaume et al., 2013; Pausas et al., 2008; Turco et al., 2017).

Most wind and bark beetle disturbances occurred in boreal and temperate regions, whereas residual activity dominated in the Mediterranean, consistent with previous findings (Patacca et al., 2022; Seidl et al., 2014; Senf & Seidl, 2021). However, the northern Mediterranean basin and the transition zone to temperate forests were more often associated with mild and rare events. Although we did not distinguish between wind- and bark beetle-driven events, previous literature suggests that bark beetle outbreaks and other biotic agents (e.g. *Fusarium circinatum* and *Thaumetopoea* spp.) are more prevalent than storm- or wind-induced disturbances in the Mediterranean and transitions to temperate forests (Forzieri et al., 2020, 2023; Patacca et al., 2022; Senf & Seidl, 2021). In boreal and temperate biomes, three distinct patterns emerged. First, moderate disturbances dominated across most regions, except in Central Europe and the British Isles, and corresponded to areas where wind is the primary disturbance agent (Schelhaas et al., 2003; Thom & Seidl, 2016). Second, small and frequent disturbances dominated in Central Europe, where wind-related disturbances are common and bark beetle outbreaks occur at comparatively higher levels than elsewhere in the biome (Patacca et al., 2022; Seidl et al., 2014; Washaya et al., 2024). This regime may not only result from frequent wind-throw and severe bark beetle outbreaks in the region but also reflect the influence of widespread salvage logging practices in managed forests across Central Europe, particularly in Germany, as well as in countries such as Czechia, Austria and Slovakia. Such practices can further clear-cut partially disturbed areas resulting in high prevalence of small disturbance patches (Leverkus et al., 2021). Third, large and severe disturbances dominated in the British Isles. While intense windstorms associated with North Atlantic depressions are frequent in this region (Forzieri et al., 2020; NEWA, 2019), bark beetles and other biotic infestations are relatively rare (Jeger et al., 2017; Romeiro et al., 2022; Seidl et al., 2014). The emergence of this cluster in the GMM may reflect the influence of two additional factors. One factor is the distinctive spatial configuration of forests in the United Kingdom and Ireland (i.e. low tree cover and small, fragmented patches) compared to temperate forests elsewhere in Europe (Köhl et al., 2020; The Research Agency of the Forestry Commission, 2024). A second factor is that some large disturbance patches may reflect the misclassification of harvesting events in plantations as wind and bark beetle disturbances in the European Atlas.

4.2 | Temporal dynamics of forest disturbances regimes

The temporal analyses revealed that nearly 15% of grid cells transitioned to different clusters between the 1985–1997 and 2011–2023 periods for both agent groups. As natural disturbances are highly stochastic, some of these transitions may have been triggered

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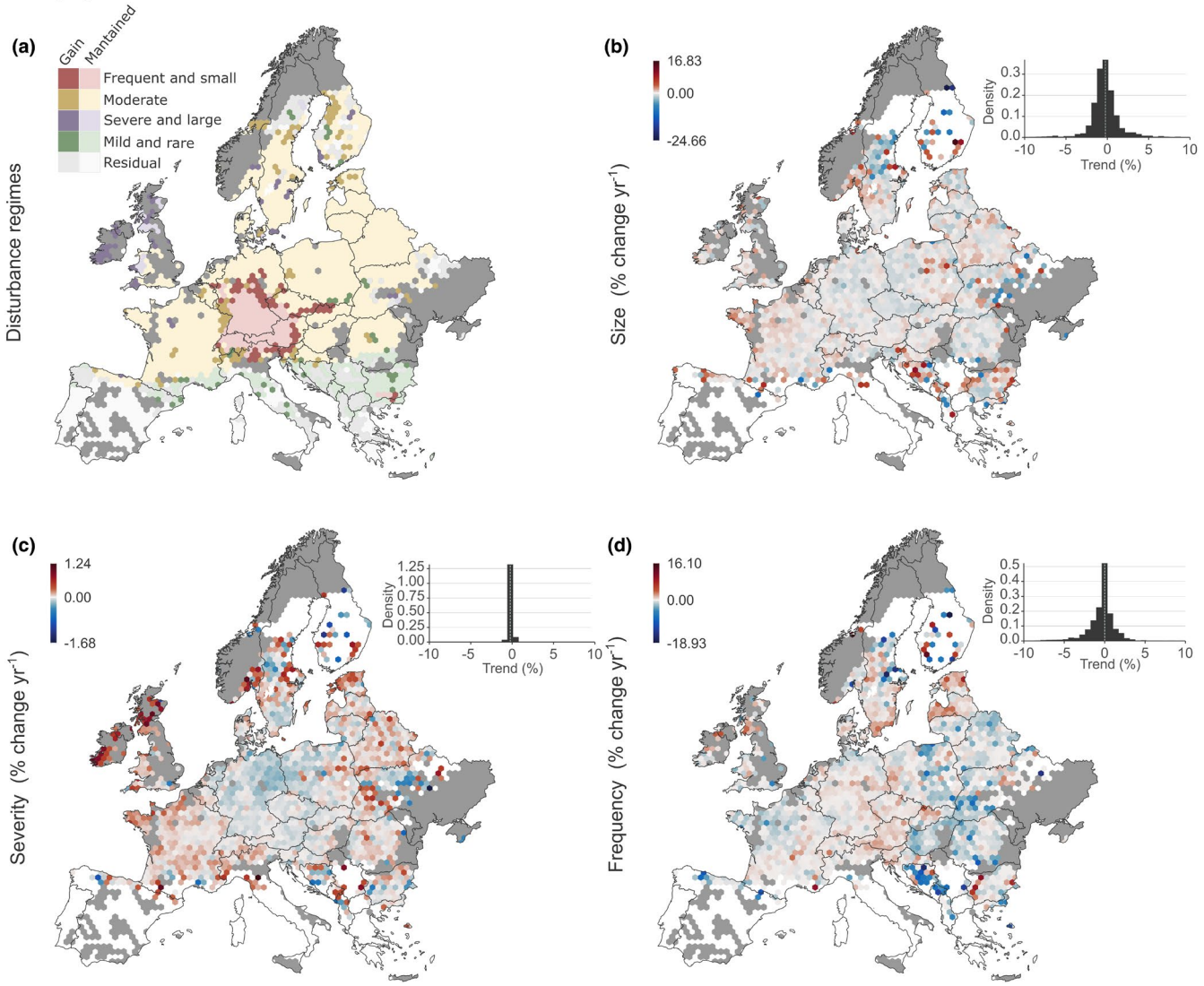


FIGURE 5 Wind and bark beetle disturbance changes through time. Panel (a) shows changes in disturbance clusters between periods first (1985–1997) and third (2011–2023). Grids classified as maintained did not change cluster regime, while gain indicates a transition to a different disturbance cluster. Panels (b–d) show relative trends (% change year^{-1}) of properties in Viana-Soto and Senf (2025) of disturbance size (b), severity (c) and frequency (d) from 1985 to 2023. Panels (b–d) display significant trends ($p < 0.05$) colouring the entire grid cells, whereas non-significant trends ($p \geq 0.05$) are shown with coloured points. Grids in grey were those excluded from the analyses.

by extreme disturbances in certain years (Senf & Seidl, 2021). Most fire-related transitions shifted towards residual activity in boreal and temperate forests, and towards severe and rare regimes in southern latitudes. These results reflect the declining trends in fire frequency and severity at the biome level in temperate and Mediterranean region, and align with earlier findings indicating a decrease in burned area and fire occurrence across much of Europe since the 1990s (Baijnath-Rodino et al., 2023; Grünig et al., 2023; Senf & Seidl, 2021; Turco et al., 2016). Despite this overall decline, substantial regional heterogeneity was found at the grid cell level, particularly in the Mediterranean regions in terms of fire size and severity. Improved fire prevention and suppression measures, increased social awareness and law-enforcement efforts, and more effective detection and response systems compared with the 1980s are among the factors contributing to this overall declining trend (Brotons et al., 2013; Ruffault & Mouillot, 2015). Nonetheless, fires are highly sensitive to weather conditions and under ongoing climate change (i.e. characterised by increased drought frequency, more frequent heat events, longer fire seasons and drier fuels), fire activity may escalate to unprecedented levels, as observed in recent fire seasons (Jones et al., 2022; Potapov et al., 2025; San-Miguel-Ayanz et al., 2019; Turco et al., 2019).

Most transitions between clusters of wind and bark beetle disturbances were towards residual activity in boreal and Mediterranean regions, and towards *frequent and small* and *severe and large* clusters in the temperate region. The expansion of *frequent and small* clusters was concentrated in Central Europe and reflects the increases in disturbance

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frequency observed in these areas. This pattern aligns with the observed spatial expansion and intensification of bark beetle outbreaks in the region (Jactel et al., 2012; Marini et al., 2017), which have been linked to increasingly warmer and drier conditions that reduce host resistance (Huang et al., 2020), facilitate bark beetle survival and development (Baier et al., 2007) and enable their expansion to higher altitudes and latitudes (Jönsson et al., 2008). However, we found no clear explanation for the negative trends in disturbance severity observed in certain parts in Central Europe. A possible explanation may be that the European Atlas may partially capture post-disturbance recovery alongside actual disturbance impacts, leading to an underestimation of severity due to temporal lags between disturbance occurrence and detection. Unlike those in Central Europe, western and eastern temperate forests generally showed non-significant or declining trends in disturbance frequency and increases in severity. This increase in severity was particularly pronounced in the British Isles, where numerous *moderate* clusters during 1985–1997 transitioned into *large and severe* clusters in 2011–2023. These temporal patterns may be driven by the increasing intensity of storms and wind-related forest damage reported since the 1990s (Gregow et al., 2017; Patacca et al., 2022), as well as by the influence of extensive forest plantations in the region. Such plantations may either contribute to larger patches of wind-related disturbances or lead to apparently more severe disturbances due to the possible misclassification of some harvest events as wind- or bark beetle-related disturbances in the European Atlas (Viana-Soto & Senf, 2025).

4.3 | Limitations

Here, we used the only available continental-scale, remote-sensing-based product that provides a detailed classification of multiple natural disturbance agents to characterise spatiotemporal variability of disturbance regimes in European forests. Although remote sensing derived products provides valuable and accurate information on disturbance events at a continental scale (Forzieri et al., 2021; Hansen et al., 2013; Viana-Soto & Senf, 2025; Vogeler et al., 2020) and have repeatedly been used to report disturbance patterns and trends (Acil et al., 2024; Senf & Seidl, 2020; Sommerfeld et al., 2018), several limitations should be considered. First, there can be a temporal lag between disturbance occurrence and detection, which may affect severity estimates reported in the European Atlas (i.e. severity may be underestimated due to early vegetation recovery captured at the time of detection; Viana-Soto & Senf, 2025). Second, although natural disturbances followed by salvage logging are classified in the European Atlas according to the root cause, post-disturbance management may modify the spatial imprint that is captured. As a result, the size, severity and frequency we analyse may partly reflect subsequent management interventions. Third, the metrics used to characterise disturbance regimes cannot fully capture the complexity of forest structure and landscape configuration. Although disturbance frequency was normalised by the forested area within each grid cell to account for differences in forest extent, this approach does not consider other aspects of landscape complexity, such as fragmentation, which may have influenced the disturbance clusters identified.

5 | MANAGEMENT APPLICATIONS AND CONCLUSIONS

The pronounced heterogeneity of disturbance regimes across Europe highlights that forests experiencing the highest disturbance activity, or the strongest changes over time, vary depending on the region, the causal agent and the attribute (i.e., size, severity, frequency). This suggests that forest management strategies should not rely on uniform approaches, but instead need to be multiscale, context-dependent and tailored to both the dominant disturbance agent and local environmental conditions.

For fires, western Mediterranean forests are facing frequent and large fires that, combined with recurrent droughts and impaired regeneration, may require priority actions that account for both long-term dynamics and future climatic conditions. Management strategies aimed at reducing fire susceptibility should adopt multiscale perspectives that integrate interventions at both stand and landscape scales. At the stand level, management should consider promoting species mixtures and stand structures that are resistant to expected future fire regimes, helping to control flammability and facilitate post-fire regeneration. Such resistant mixtures may include oak species with thick bark and strong resprouting capacity, as well as Mediterranean serotinous species with high post-fire seeding capacity (Barrere et al., 2023; Keeley et al., 2011; Keeley & Pausas, 2022). In dense forests, excessive biomass could be reduced through selective cutting or herbivory, particularly where regeneration processes are not compromised. At the landscape level, priority strategies should focus on managing horizontal and vertical fuel continuity to enhance landscape heterogeneity and reduce potential fire size and severity.

Central Europe is experiencing pronounced increases in wind and bark beetle disturbance frequency. Management actions in these critical regions may prioritise reducing large, continuous patches of host species for *Ips typographus* (e.g. *Picea abies* L.) and promoting heterogeneous stands through the admixture of other tree species (e.g. *Fagus sylvatica* L. or *Abies alba* Mill.; Seidl et al., 2016). These actions should also account for both current and projected environmental conditions, and favour species with structural traits better adapted to future climates, such as dense-wooded and larger stature species that are more resistant to both storms and bark beetle attacks (Barrere et al., 2024). In addition, an improved understanding of disturbance dynamics is particularly needed in critical transition zones, such as in the Mediterranean–temperate transition region, where moderate fire activity coexists with wind-related disturbances and increasing bark beetle outbreaks, making linked or interacting disturbances more likely (Buma, 2015).

Our results revealed that disturbance regimes across Europe are highly dynamic, with nearly 15% of forests shifting between disturbance regimes over the study period. Changes in disturbance regimes pose an increasing threat to forest ecosystems, although this threat is strongly

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context dependent: fire regimes are changing predominantly in the Mediterranean Basin, whereas wind and bark beetle disturbances are changing more in temperate forests. These shifts should be considered together with broader climate change trends, including warming and altered water availability, to better understand potential impacts on forest resilience through increased mortality, reduced growth or altered regeneration (McDowell et al., 2020).

AUTHOR CONTRIBUTIONS

Sofia Miguel, Emily Lines and Paloma Ruiz-Benito conceived the ideas and designed methodology; Sofia Miguel analysed and interpreted the data; Sofia Miguel and Paloma Ruiz-Benito led the writing of the manuscript. Emily Lines, Paloma Ruiz-Benito, Alba Viana-Soto, Mihai Tanase and Cornelius Senf reviewed and edited different drafts of the manuscript. All authors contributed critically to the drafts and gave final approval for publication.

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CONFLICT OF INTEREST STATEMENT

The authors have no conflicts of interest to disclose.

DATA AVAILABILITY STATEMENT

Data and code available are from the Dryad Digital Repository: https://doi.org/10.5061/dryad.gb5mkkx4q (Miguel et al., 2026).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Figure S1. Histograms showing the mean ΔNBR (used here as the severity metric) of all disturbed patches considered in the GMM analysis, before and after scaling.

Figure S2. Correlation plot of candidate variables considered for the Gaussian finite mixture model (GMM) analyses, for (A) fire and (B) wind and bark beetle.

Figure S3. Number of grid cells removed using different thresholds of minimum disturbance events for each disturbance agent.

Figure S4. Spatial distribution of disturbance clusters for fire (A, B) and wind and bark beetle (C, D).

Figure S5. Spatial distribution of disturbance clusters for fire (A, B) and wind and bark beetle (C, D), obtained from GMM analyses using two grid cell sizes: 25 km (A, C) and 100 km (B, D).

Figure S6. Spatial distribution of disturbance clusters for fire (A, B, C) and wind and bark beetle (D, E, F) obtained from GMM analyses.

Table S1. Summary of disturbance attributes driven by fire and wind and bark beetle disturbances across the periods 1985–1997, 1998–2010 and 2011–2023.

Figure S7. Proportion (%) of grid cells in each disturbance cluster for fire (A–C) and wind and bark beetle (D–F) across the periods 1985–1997 (P1), 1998–2010 (P2) and 2011–2023 (P3), within the boreal (A, D), temperate (B, E) and Mediterranean (C, F).

Figure S8. Number of grid cell transitions between disturbance clusters from the period 1985–1997 to the period 2011–2023 for (A) fire and (B) wind and bark beetle disturbances.

Table S2. Comparison of Gaussian finite mixture models fitted using different sets of three disturbance variables and different scales.

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